

Phosphorus, Total

DOC316.53.01123

Molybdovanadate Method with Acid Persulfate Digestion¹

Method 10127

HR (1.0 to 100.0 mg/L PO₄³⁻)

Test 'N Tube™ Vials

Scope and Application: For water and wastewater

¹ Adapted from *Standard Methods for the Examination of Water and Wastewater.*, (4500 B-C)

Test preparation

How to use instrument-specific information

The [Instrument-specific information](#) table displays requirements that may vary between instruments. To use this table, select an instrument then read across to find the corresponding information required to perform this test.

Table 367 Instrument-specific information

Instrument	Light shield
DR 6000	—
DR 5000	—
DR 3900	LZV849
DR 3800, DR 2800, DR 2700	LZV646

Before starting the test:

DR 3900, DR 3800, DR 2800 and DR 2700: Install the light shield in Cell Compartment #2 before performing this test.

Reagent blanks can be used more than once, but should not be used more than one day.

The final samples will contain molybdenum. In addition, the final samples will have a pH less than 2 and are considered corrosive (D002) by the Federal RCRA. Refer to the current MSDS for safe handling and disposal instructions.

Collect the following items:

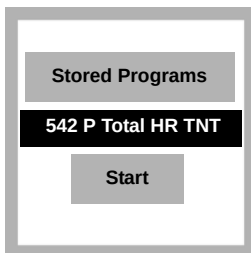
Description	Quantity
Total High Range Phosphorus Test 'N Tube™ Reagent Set	1
DRB200 Reactor, 15 x 16 mm	1
Light Shield or Adapter (see Instrument-specific information)	1
Pipet, TenSette®, 1 to 10 mL, plus tips	1
Test Tube Rack	1
Funnel, Micro	1

See [Consumables and replacement items](#) for reorder information.

Molybdovanadate method, acid persulfate digestion



1. Turn on the DRB200 Reactor. Heat to 150 °C.



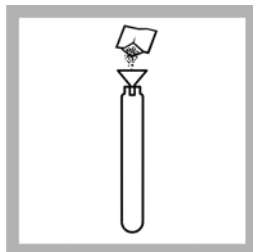
2. Select the test. Insert an adapter if required (see [Instrument-specific information](#)).



3. **Blank Preparation:** Use a TenSette® Pipet to add 5.0 mL of deionized water to a Total Phosphorus Test 'N Tube Vial.



4. **Prepared Sample:** Use a TenSette Pipet to add 5.0 mL of sample to a Total Phosphorus Test 'N Tube Vial.



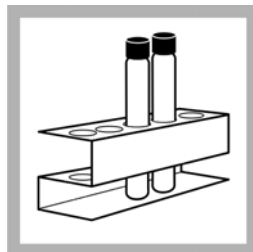
5. Use a funnel to add the contents of one Potassium Persulfate Powder Pillow to each vial. Cap tightly and shake to dissolve.



6. Insert the vials in the DRB 200 Reactor. Close the protective cover.

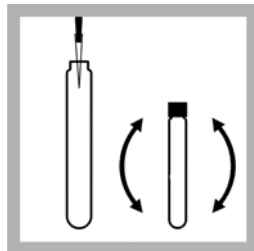


7. Set the instrument timer to 30 minutes and start the timer. A 30-minute heating period will begin.



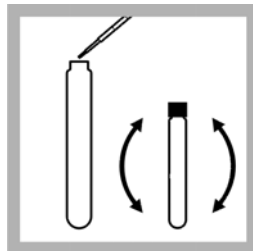
8. After the timer expires, carefully remove the hot vials from the reactor. Insert them in a test tube rack and allow to cool to room temperature (18–25 °C).

Molybdovanadate method, acid persulfate digestion (continued)



9. Use a TenSette Pipet to add 2.0 mL of 1.54 N sodium hydroxide to each vial.

Cap and invert to mix.



10. Use a polyethylene dropper to add 0.5 mL of Molybdovanadate Reagent to each vial.

Cap and invert to mix.



11. Start the instrument timer.

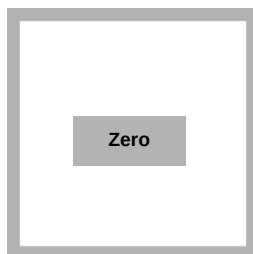
A 7-minute reaction period will begin. Read the sample between seven and nine minutes after adding the Molybdovanadate Reagent.



12. Wipe the vials with a damp towel, followed by a dry one, to remove fingerprints or other marks.



13. When the timer expires, insert the blank into the 16-mm cell holder.



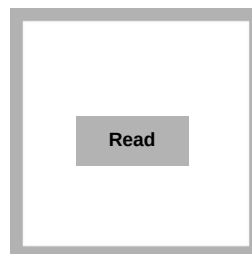
14. ZERO the instrument.

The display will show:

0.0 mg/L PO_4^{3-}



15. Insert the prepared sample into the 16-mm cell holder.



16. READ the results in mg/L PO_4^{3-} .

Interferences

Sample turbidity may cause inconsistent results in the test because the acid present in the reagents may dissolve some of the suspended particles and because of variable desorption of orthophosphate from the particles.

The [Interfering substances](#) table shows interference levels and types. The [Noninterfering substances, \(at concentrations less than 1000 mg/L\)](#) table shows substances that do not interfere in concentrations less than 1000 mg/L.

Table 368 Interfering substances

Interfering substance	Interference level
Arsenate	Causes positive interference if the sample is warm when the molybdovanadate reagent is added (after the digestion). ¹ Cool the sample after digestion before adding reagent.
Iron, ferrous	Blue color caused by ferrous iron does not interfere if iron concentration is less than 100 mg/L.

Table 368 Interfering substances (continued)

Interfering substance	Interference level
Molybdate	Causes negative interference above 1000 mg/L.
Silica	Causes positive interference if the sample is warm when the molybdovanadate reagent is added (after the digestion). Cool the sample after digestion before adding reagent.
Extreme pH or highly buffered samples	May exceed buffering capacity of the reagents. Samples may require pretreatment. Sample pH should be about 7.
Fluoride, thorium, bismuth, thiosulfate or thiocyanate	Cause a negative interference.
Temperature, Cold (less than 18 °C)	Cause a negative interference.
Temperature, Hot (greater than 25 °C)	Causes a positive interference. Post-digestion samples should be brought to room temperature (18–25 °C) before the addition of the Molybdovanadate Reagent or sodium hydroxide.

¹ Gentle warming of the sample to reach room temperature will not cause this substance to interfere.

Table 369 Noninterfering substances, (at concentrations less than 1000 mg/L)

Pyrophosphate	Tetraborate	Selenate	Benzoate
Citrate	Oxalate	Lactate	Tartrate
Formate	Salicylate	Al ³⁺	Fe ³⁺
Mg ²⁺	Ca ²⁺	Ba ²⁺	Sr ²⁺
Li ⁺	Na ⁺	K ⁺	NH ₄ ⁺
Cd ²⁺	Mn ²⁺	NO ₃ ⁻	NO ₂ ⁻
SO ₄ ²⁻	SO ₃ ²⁻	Pb ²⁺	Hg ⁺
Hg ²⁺	Sn ²⁺	Cu ²⁺	Ni ²⁺
Ag ⁺	U ⁴⁺	Zr ⁴⁺	AsO ₃ ⁻
Br ⁻	CO ₃ ²⁻	ClO ₄ ⁻	CN ⁻
IO ₃ ⁻	SiO ₄ ⁴⁻	—	

Sample collection, preservation and storage

- Collect samples in plastic or glass bottles that have been acid washed with 1:1 Hydrochloric Acid Solution* and rinsed with deionized water.
- Do not use commercial detergents containing phosphate for cleaning the glassware used in this test.
- Analyze samples immediately after collection for best results.
- If prompt analysis is impossible, preserve samples up to 28 days by adjusting the pH to 2 or less with concentrated Sulfuric Acid* (about 2 mL per liter) and storing at 4 °C.
- Warm stored samples to room temperature and neutralize with 5.0 N Sodium Hydroxide* before analysis.
- Correct the test results for volume additions.

* See [Optional reagents and apparatus](#).

Accuracy check

Standard additions method (sample spike)

Required for accuracy check:

- 10-mL Voluette® Ampule of Phosphate Standard Solution, 500 mg/L as PO_4^{3-}
 - Ampule breaker
 - TenSette Pipet and tips
 - Mixing cylinders (3)
1. Clean glassware with 1:1 Hydrochloric Acid Standard Solution. Rinse again with deionized water. Do not use detergents containing phosphate to clean glassware.
 2. After reading test results, leave the sample cell (unspiked sample) in the instrument.
 3. Select standard additions from the instrument menu **OPTIONS>MORE>STANDARD ADDITIONS**
 4. Accept the default values for standard concentration, sample volume and spike volumes. After the values are accepted, the unspiked sample reading will appear in the top row. See the user manual for more information.
 5. Open the standard solution ampule.
 6. Use the TenSette Pipet to prepare spiked samples: add 0.1 mL, 0.2 mL and 0.3 mL of standard to three 10-mL portions of fresh sample in 25 mL mixing cylinders.
 7. Follow the [Molybdovanadate method](#), [acid persulfate digestion](#) test procedure for each of the spiked samples, starting with the 0.1 mL sample spike. Measure each of the spiked samples in the instrument.
 8. Select **GRAPH** to view the results. Select **IDEAL LINE** (or best-fit) to compare the standard addition results to the theoretical 100% recovery.

Standard solution method

Note: Refer to the instrument user manual for specific software navigation instructions.

Required for accuracy check:

- Phosphate standard solution, 50-mg/L
- 1. Use the 50-mg/L phosphate standard solution in place of the sample. Follow the [Molybdovanadate method](#), [acid persulfate digestion](#) test procedure.
- 2. To adjust the calibration curve using the reading obtained with the standard solution, navigate to Standard Adjust in the software **OPTIONS>MORE>STANDARD ADJUST**.
- 3. Turn on the Standard Adjust feature and accept the displayed concentration. If an alternate concentration is used, enter the concentration and adjust the curve to that value.

Method performance

Program	Standard	Precision—95% Confidence Limits of Distribution	Sensitivity— Δ Concentration per 0.010 Δ Abs
542	50 mg/L PO_4^{3-}	49.4–50.6 mg/L PO_4^{3-}	0.7 mg/L PO_4^{3-}

Summary of method

Phosphates present in organic and condensed inorganic forms (meta-, pyro- or other polyphosphates) must be converted to reactive orthophosphate before analysis. Pretreatment of the sample with acid and heat provides the conditions for hydrolysis of the condensed inorganic forms. Organic phosphates are converted to orthophosphates by heating with acid and persulfate.

Orthophosphate reacts with molybdate in an acid medium to produce a mixed phosphate/molybdate complex. In the presence of vanadium, yellow molybdovanadophosphoric acid forms. The intensity of the yellow color is proportional to the phosphate concentration. Test results are measured at 420 nm.

Consumables and replacement items

Required reagents

Description	Quantity/Test	Unit	Catalog number
Total High Range Phosphorus Test 'N Tube™ Reagent Set, includes:	—	50 vials	2767245
(1) Molybdovanadate Reagent ¹	0.5 mL	25 mL	—
(1) Potassium Persulfate powder Pillows	1	50/pkg	2084766
(1) Sodium Hydroxide Solution, 1.54 N	2 mL	100 mL	2743042
(1) Total Phosphorus Test Vials ¹	1	50/pkg	—
(2) Water, deionized	5 mL	100 mL	27242

¹ Not available separately.

Required apparatus

Description	Quantity	Unit	Catalog number
DRB200 Reactor, 110 V, 15 x 16 mm	1	each	LTV082.53.40001
DRB200 Reactor, 220 V, 15 x 16 mm	1	each	LTV082.52.40001
Dropper, 0.5 and 1.0 mL marks, plastic	1	20/pkg	2124720
Light shield, DR 3900	1	each	LZV849
Light shield, DR 3800, DR 2800, DR 2700	1	each	LZV646
Pipet, TenSette®, 1 to 10 mL	1	each	1970010
Pipet Tips, for TenSette Pipet 19700-10	1	250/pkg	2199725
Test Tube Rack	1	each	1864100
Funnel, micro	1	each	2584335

Recommended standards

Description	Unit	Catalog number
Phosphate Standard Solution, Voluette® ampule, 500-mg/L as PO_4^{3-} , 10-mL	16/pkg	1424210
Phosphate Standard Solution, 50-mg/L as PO_4^{3-}	500 mL	17149
Wastewater Standard, Influent Inorganics for $\text{NH}_3\text{-N}$, $\text{NO}_3\text{-N}$, PO_4 , COD, SO_4 , TOC	500 mL	2833149
Voluette Ampule breaker 10 mL	each	2196800

Optional reagents and apparatus

Description	Unit	Catalog number
Hydrochloric Acid Solution, 6.0 N, 1:1	500 mL	88449
Sodium Hydroxide, 5.0 N	1000 mL	245053
Sulfuric Acid, concentrated	500 mL	97949
Molybdovanadate Reagent	100 mL	2076032
Pipet Tips, for TenSette Pipet 1970010 ¹	50/pkg	2199796
Pipet, TenSette®, Pipet, 0.1–1.0 mL	each	1970001
Pipet Tips, for TenSette Pipet 1970001 ¹	50/pkg	2185696

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Optional reagents and apparatus (continued)

Description	Unit	Catalog number
Pipet Tips, for TenSette Pipet 1970001 ¹	1000/pkg	2185628
pH Paper, 0–14 pH range	100/pkg	2601300
Deionized Water	4 L	27256
Thermometer, Non-Mercury, -10 to 225 °C	each	2635700
Finger cots	2/pkg	1464702
Sampling Bottle with cap, low density polyethylene, 250 mL	12/pkg	2087076
Cylinder, mixing, 25 mL	each	189640
Funnel, micro	each	2584335

¹ Other sizes are available

Optional standards

Description	Unit	Catalog number
Phosphate, Standard Solution, 30 mg/L	946 mL	1436716
Phosphate, Standard Solution, 100 mg/L	100 mL	1436832
Phosphate, Standard Solution, 50 mg/L, 10 mL Voluette Ampules	16/pkg	17110
Phosphate, Standard Solution, 500 mg/L	100 mL	1424232



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